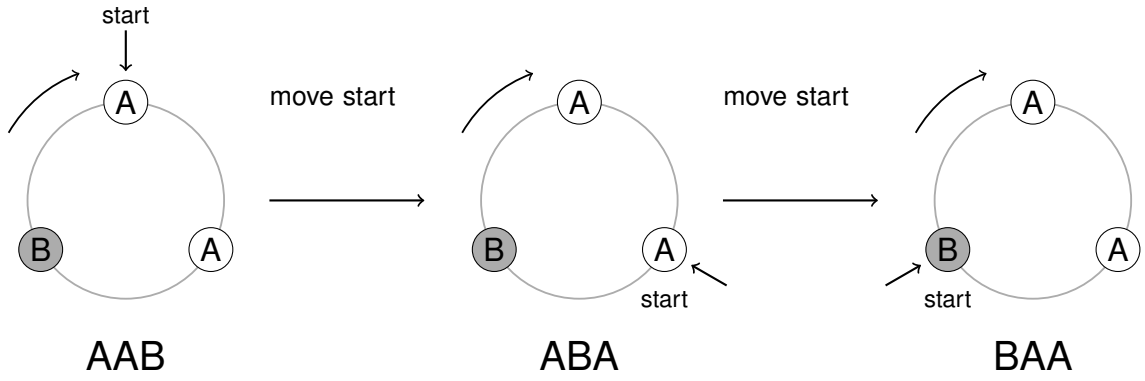
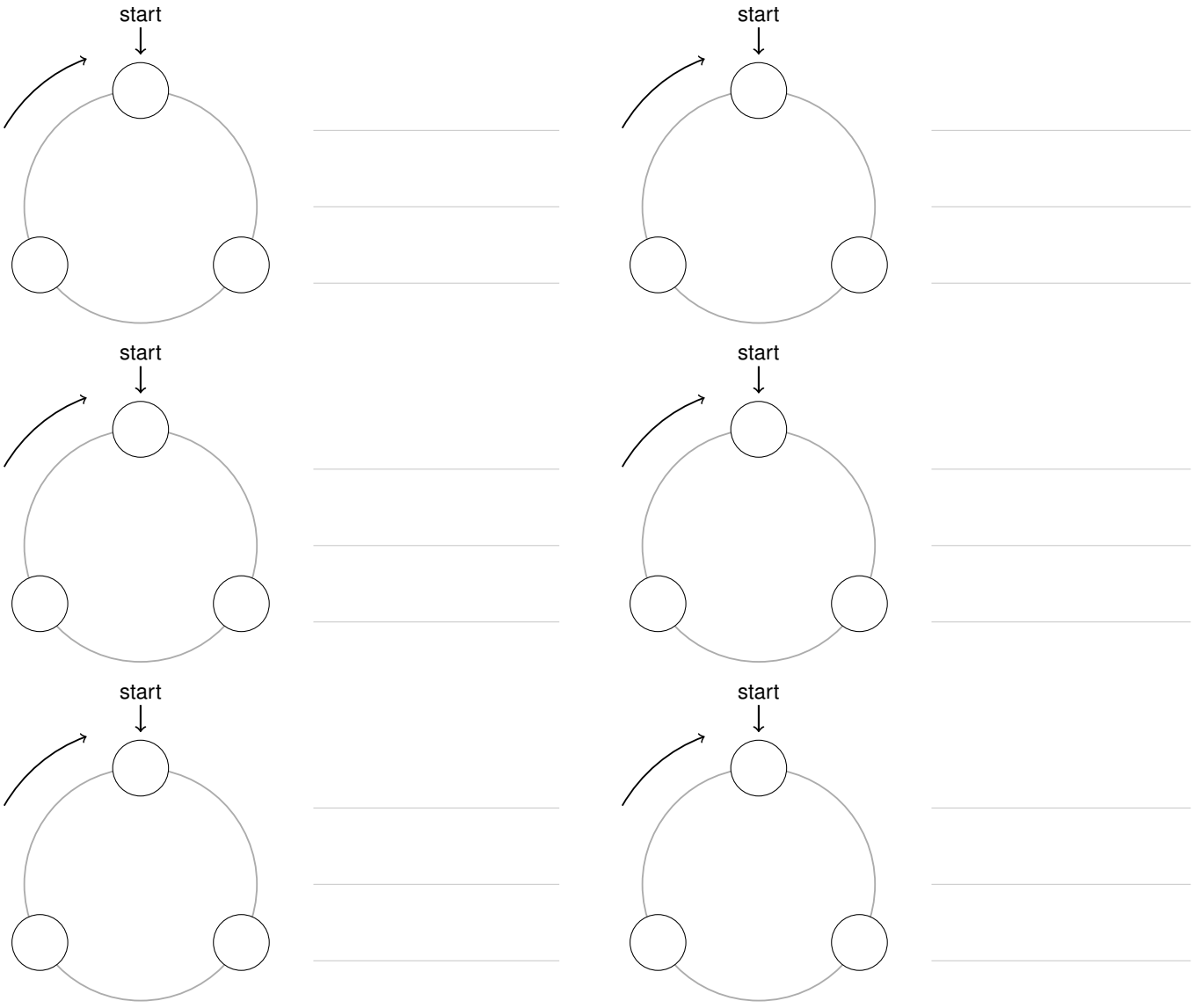
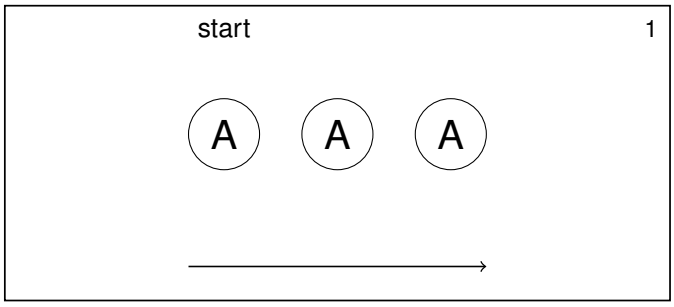
Each bead may be A or B; one-color rings are allowed. Read clockwise from the start arrow. The bead order is the readout. Rings count as the same when turning makes them match. Keep them face up; do not flip them over.

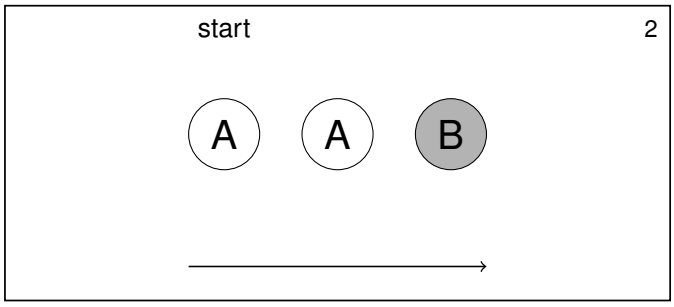


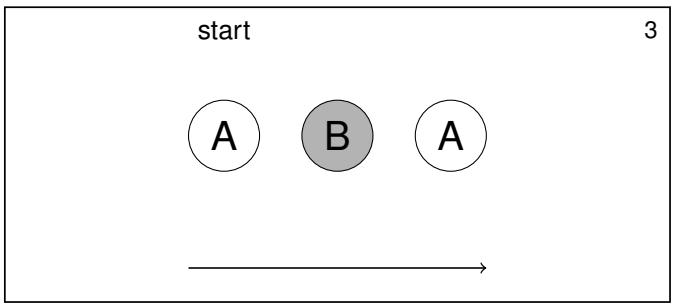
Problem 1: Make every different three-bead ring using A and B.

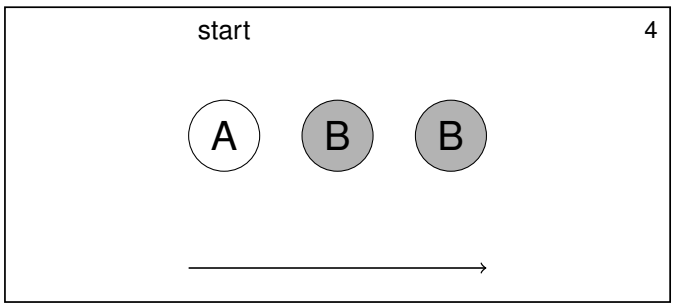


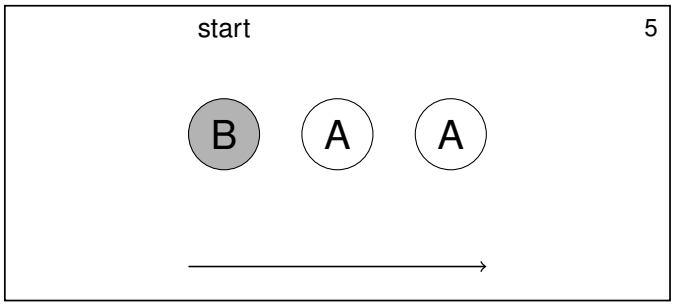
Problem 2: Put cards in the same group when they make the same ring. Do all groups have the same number of cards?

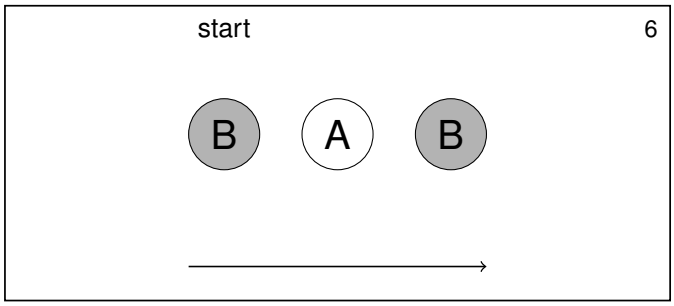


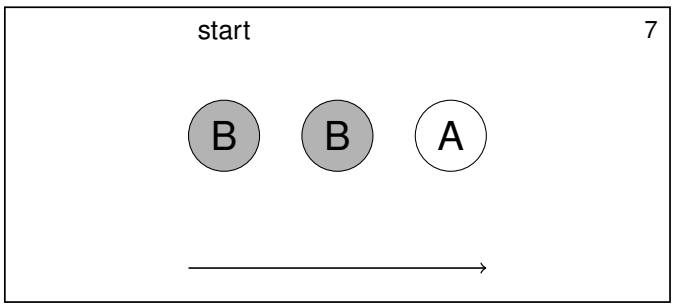


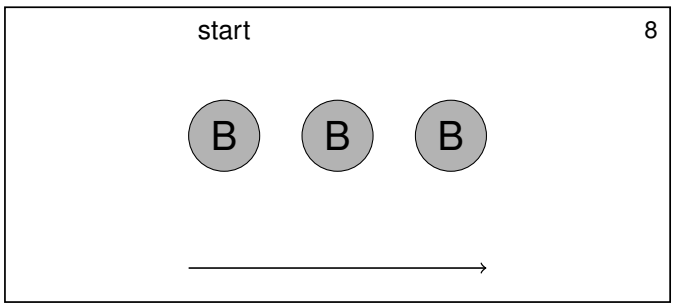




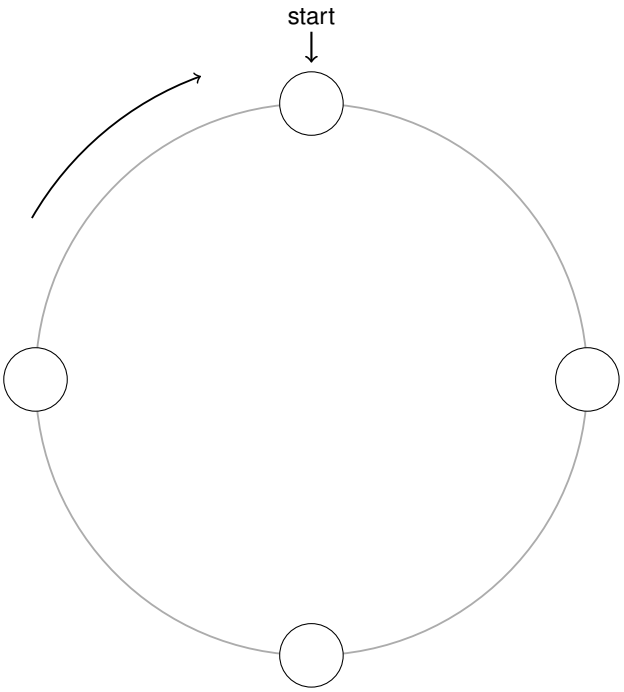


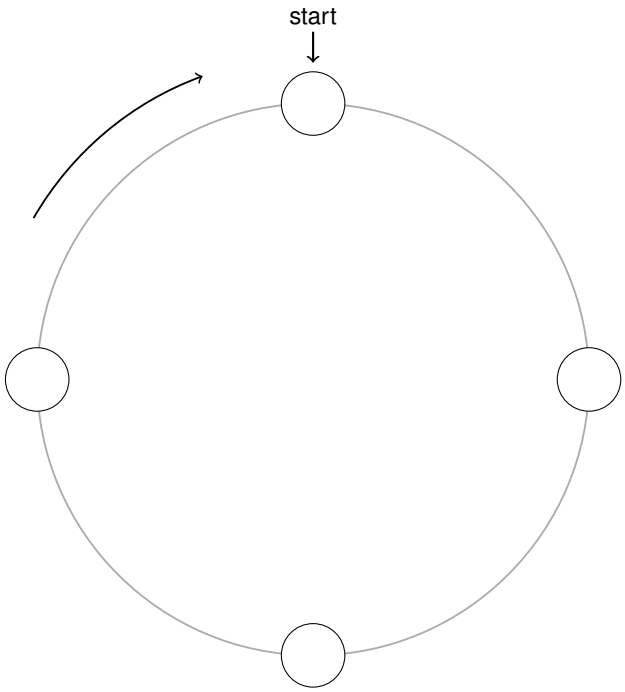


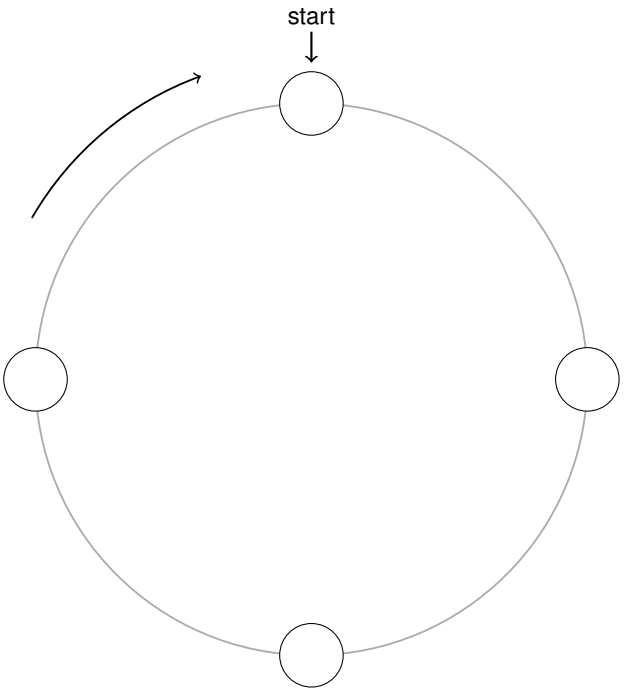


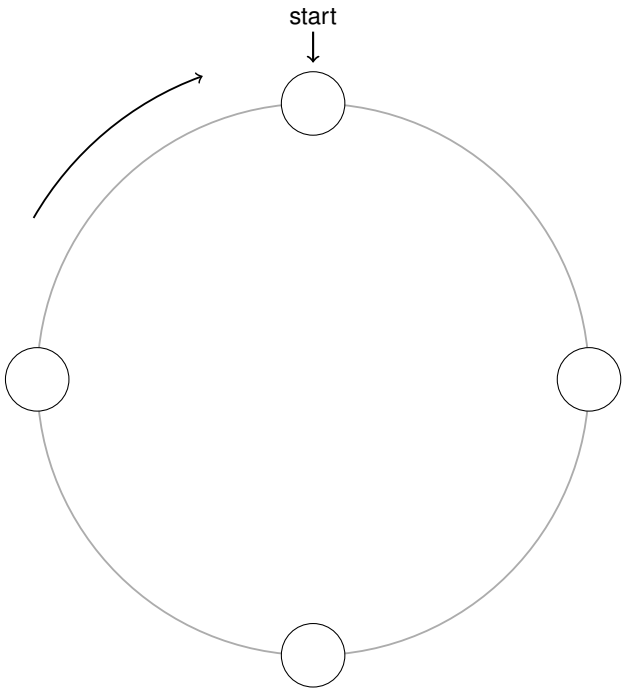


Problem 3: Find every different four-bead ring with two A beads and two B beads.

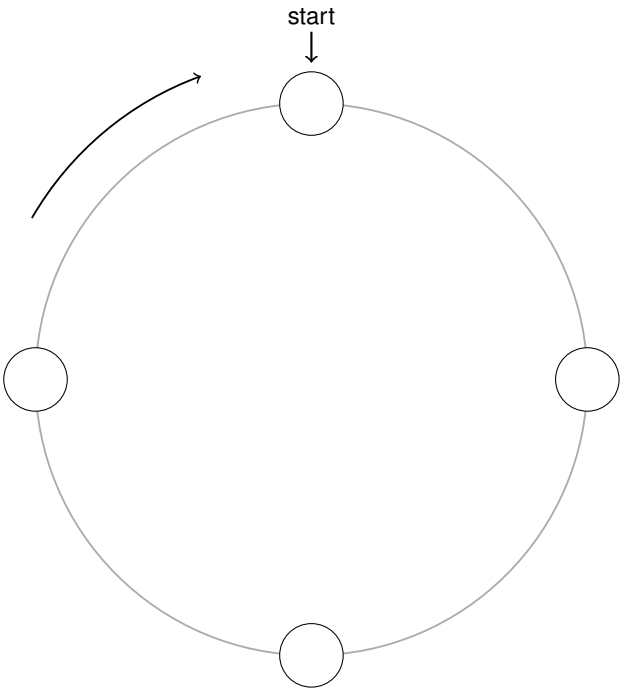


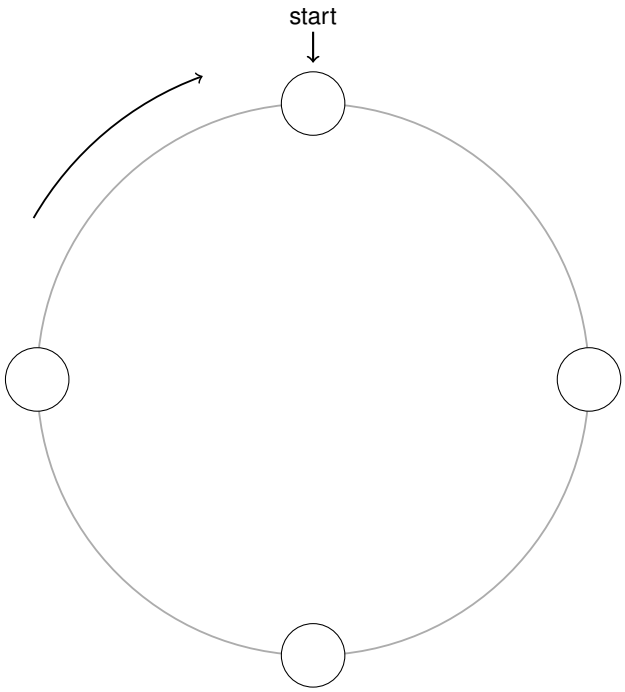


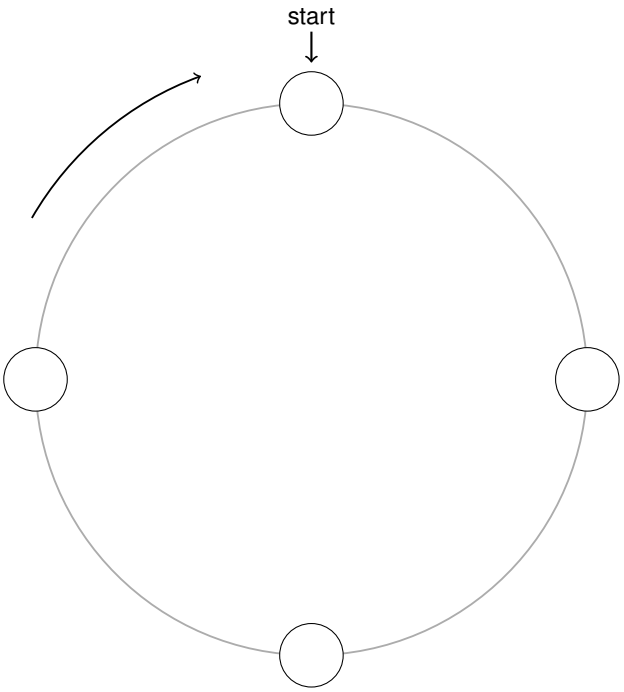


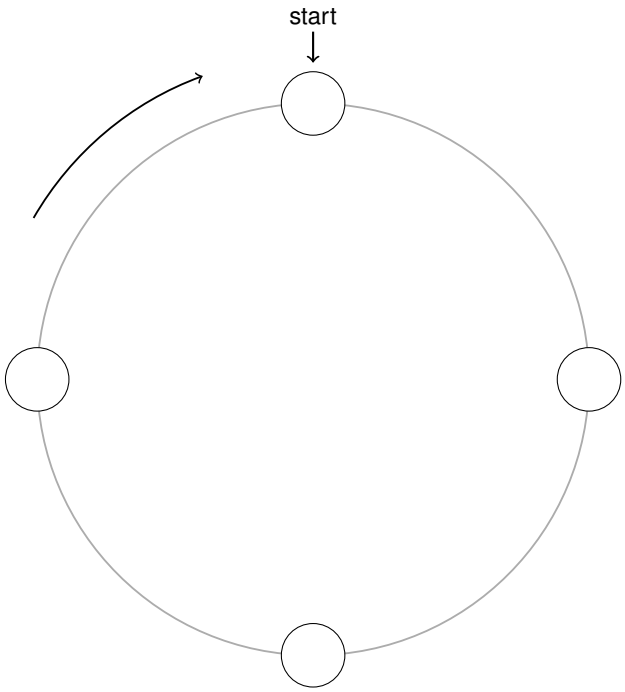


Problem 4: Make four-bead rings with one, two, three or four different readouts. Which are possible?

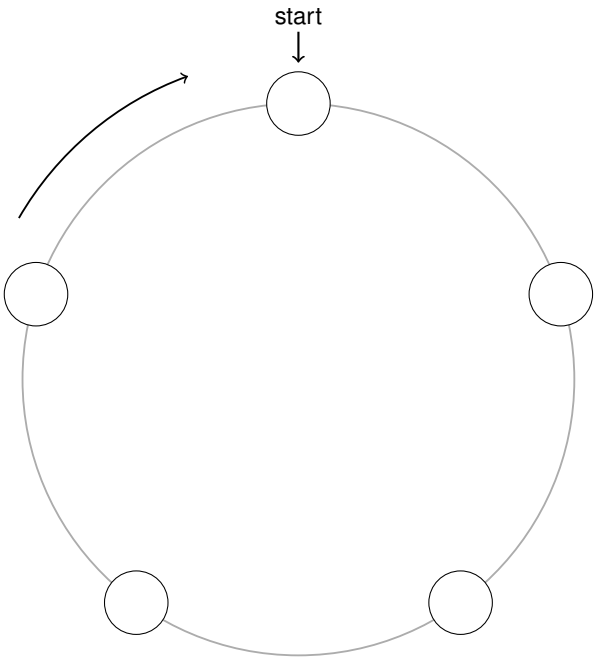


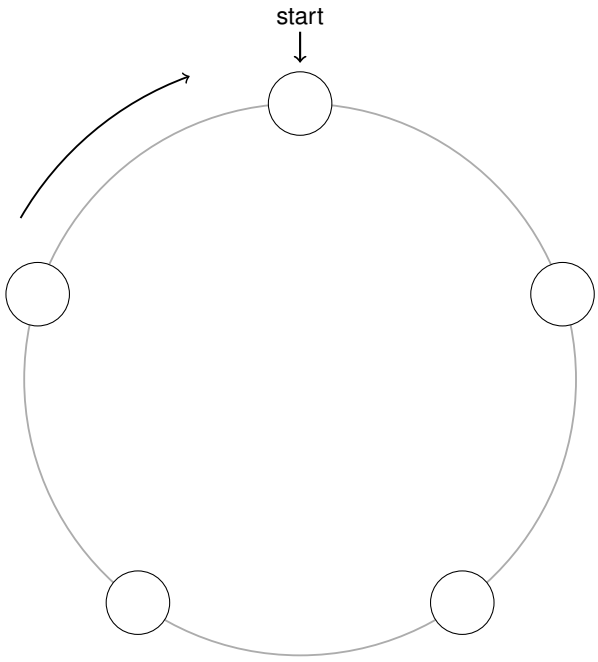


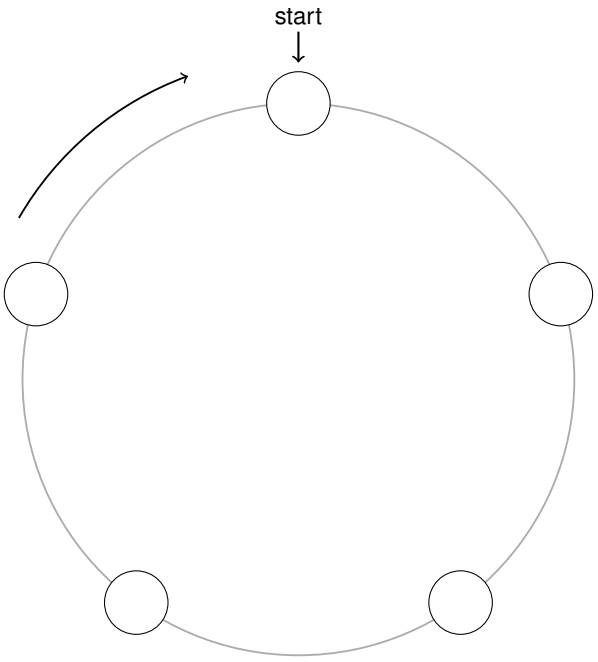


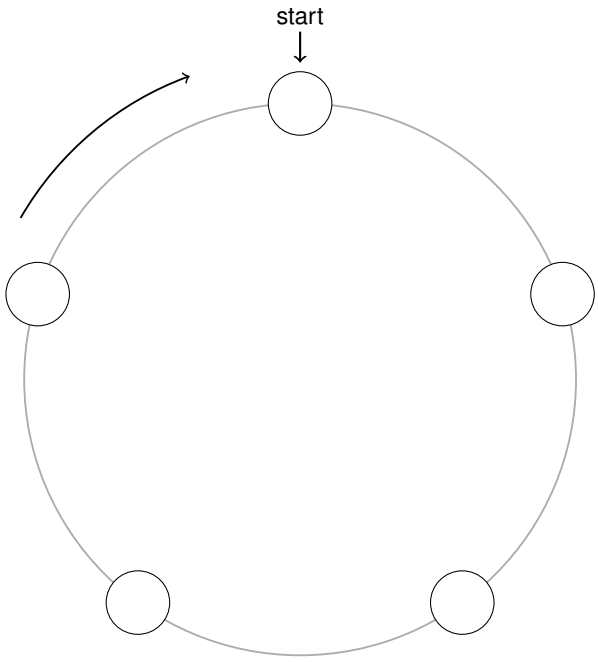
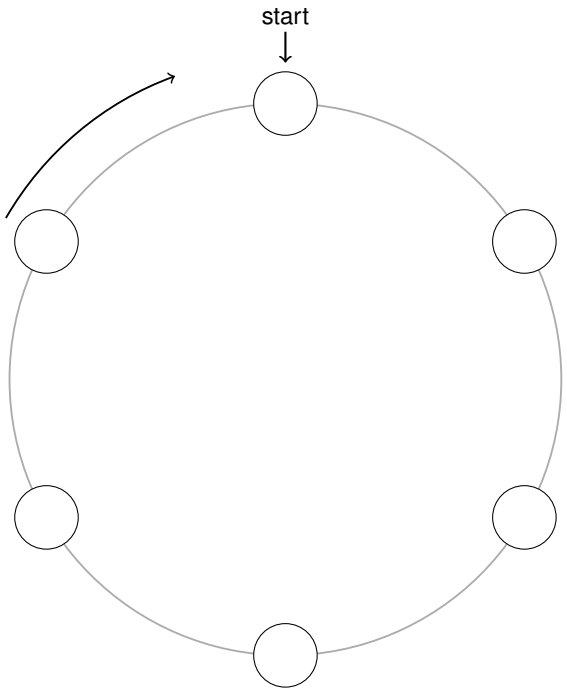
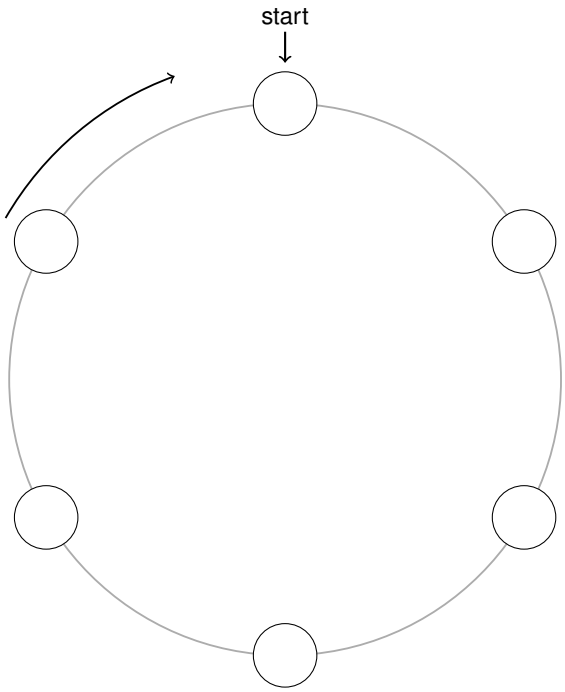
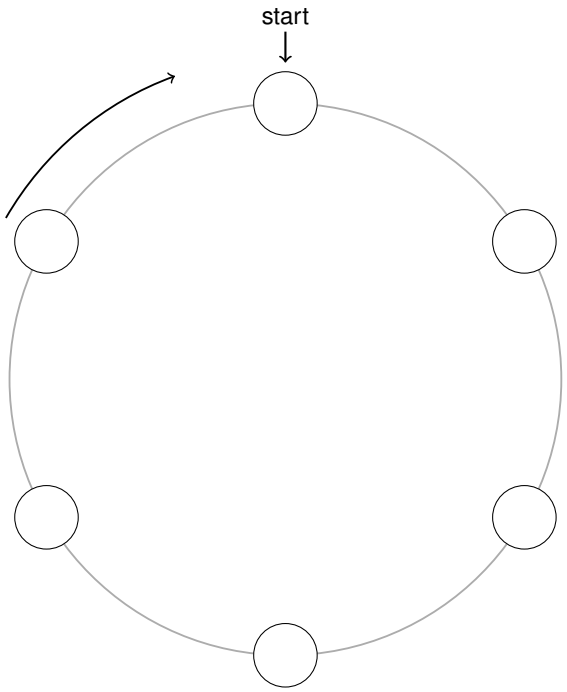
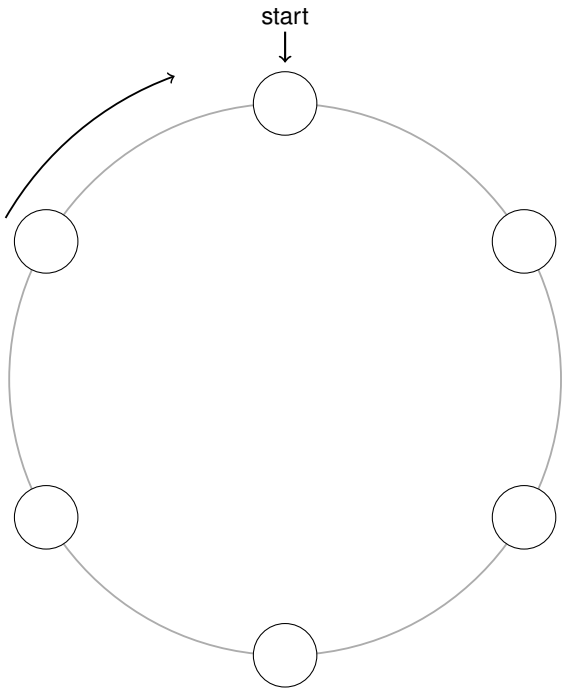


Problem 5: Find every different five-bead ring with two A beads and three B beads. How many different readouts does each have?









Problem 6: Use both A and B on a six-bead ring. Find rings with two, three and six different readouts.